

NVIDIA[®] OptiX[™] Ray Tracing SDK

Release Notes

Version 7.7.0

March 2023

Welcome to the 7.7.0 release of the NVIDIA OptiX SDK. This release adds support for Displacement Micromaps (DMMs) that provide hardware accelerated displacement for triangle mesh geometry. DMMs are supported on GPUs with RTCores and accelerated in hardware on the new Ada Lovelace generation of GPUs. Cubic curves can now be specified with a Bezier basis, and we introduce support for quadratic B-Spline ribbons (curves with a flat cross-section). The denoiser now supports temporal denoising in conjunction with 2x upscaling.

Upgrading to 7.7.0 will require source code changes, though they should be minimal. Applications compiled with the 7.7.0 SDK headers will require driver version 530 or later. Applications compiled with earlier SDK headers will continue to work, but recompiling with 7.7.0 will expose new features and may also improve performance.

System Requirements

(for running binaries referencing NVIDIA OptiX)

Graphics Hardware:

- All NVIDIA GPUs of Compute Capability 5.0 (Maxwell) or higher are supported

Graphics Driver:

- NVIDIA OptiX 7.7.0 requires that you install an R530+ driver
- Windows 8.1/10 64-bit; Linux RHEL 4.8+ or Ubuntu 10.10+ 64-bit

CUDA Toolkit

- It is **not** required to have any CUDA toolkit installed to run NVIDIA OptiX-based applications

Development Environment Requirements

(for compiling with NVIDIA OptiX)

- CUDA Toolkit 12.0

This release has been tested with OptiX IR and PTX generated from CUDA Toolkit 12.0. Other toolkit versions should also work, but 12.0 is recommended.

Version 11.7 of the CUDA toolkit introduces the OptiX IR target with `--optix-ir` (see debugging, below).

- C/C++ Compiler

A compiler compatible with the used CUDA Toolkit is required. Please see the CUDA Toolkit documentation for more information on supported compilers.

What's new in this version

- New feature: Displacement MicroMaps (DMMs)
 - DMMs provide hardware-accelerated displacement for triangle mesh geometry, turning triangles into a displaced micro-mesh. DMMs are supported on all GPUs with RTCores and accelerated in hardware on the new Ada Lovelace generation of GPUs. DMMs describe the scalar displacement values in a barycentric grid and are stored compactly with low precision and optional compression.
 - DMMs are stored in a new resource, the DMM array, which needs to be built similar to acceleration structures. DMMs in a DMM array can have different subdivision levels and formats.
 - Each `OptixBuildInputTriangleArray` can now specify an `OptixBuildInputDisplacementMicromap`. The build input references a DMM array and individual triangles in the `OptixBuildInputTriangleArray` index into the DMM array to pick the required DMM. Per vertex displacement directions and the DMM allow turning a triangle into displaced micro-mesh triangle primitive.
 - Details can be found in the OptiX programming guides, section “Displaced micro-meshes”
 - New host functions for creating arrays of displacement micromaps
 - `optixDisplacementMicromapArrayComputeMemoryUsage`
 - `optixDisplacementMicromapArrayBuild`
 - New device functions
 - `optixGetMicroTriangleVertexData`
 - `optixGetMicroTriangleBarycentricsData`
 - `optixIsDisplacedMicromeshTriangleHit`
 - `optixIsDisplacedMicromeshTriangleFrontFaceHit`
 - `optixIsDisplacedMicromeshTriangleBackFaceHit`
 - `optixBaseBarycentricsToMicroBarycentrics` - found in `optix_micromap.h`
 - New `OptixPrimitiveType`
 - `OPTIX_PRIMITIVE_TYPE_DISPLACED_MICROMESH_TRIANGLE`
- Now that OptiX can accept both PTX and OptiX-IR as input, API names were simplified
 - `optixModuleCreateFromPTX` -> `optixModuleCreate`
 - `optixModuleCreateFromPTXWithTasks` -> `optixModuleCreateWithTasks`
 - `OPTIX_ERROR_INVALID_PTX` -> `OPTIX_ERROR_INVALID_INPUT`
- `optixProgramGroupGetStackSize` now takes an optional argument for the `OptixPipeline`
 - When the pipeline parameter is non-null, OptiX will calculate the stack size based on the linked result. If OptiX programs call functions in other modules, this will produce a more accurate stack size.
- New function `optixAccelEmitProperty`
 - Emits a single property after an acceleration structure was built
 - You may still emit properties as part of `optixAccelBuild`
- `rayFlags` to `optixTrace` is now 16 bits instead of 8 bits
- Curves
 - New curve types
 - `OPTIX_PRIMITIVE_TYPE_FLAT_QUADRATIC_BSPLINE` (ribbons)
 - `OPTIX_PRIMITIVE_TYPE_ROUND_CUBIC_BEZIER`
 - Added new device functions to access curve parameters and other curve properties at the current intersection point. It is recommended that these be used instead of interpolating directly from the attributes. Recommended usage is shown in sample `optixCurves`.
 - `optixGetCubicBezierVertexData`

- `optixGetRibbonVertexData`
 - `optixGetRibbonNormal`
 - `optixGetRibbonParameters`
 - In order to ensure compatibility with future versions of OptiX, use `optixGetCurveParameter` instead of accessing the attribute directly
- `OptixInstance::sbtOffset` is now summed across all instances in the traversal graph instead of taking only the bottom-most sbt offset. This allows for sbt paletting at any particular instance level in a scene graph with multiple levels of instances (instead of only the bottom most instance level referencing the GAS), or recursive sbt paletting at each instance level in a scene graph.
- `OPTIX_BUILD_FLAG_PREFER_FAST_TRACE` is now mutually exclusive with `OPTIX_BUILD_FLAG_PREFER_FAST_BUILD`. Previously `FAST_TRACE` would override `FAST_BUILD`.
- Denoiser
 - Added new pixel formats for input guide layers
 - `OPTIX_PIXEL_FORMAT_HALF1`
 - `OPTIX_PIXEL_FORMAT_FLOAT1`
 - Added new model
 - `OPTIX_DENOISER_MODEL_KIND_TEMPORAL_UPSCALE2X`
 - Added new guide layer
 - `flowTrustworthiness`
 - Added `OptixDenoiserAOVType` to the `OptixDenoiserLayer`
- Removed `OptixPipelineLinkOptions::debugLevel` as it doesn't do anything and is unnecessary
- New and updated SDK samples
 - Headers have been rearranged to avoid the need of `optix_7_*` headers. No code changes should be expected to accommodate this change.
 - New sample `optixDisplacedMicroMesh`
 - Demonstrates use of displaced micro meshes to display finer grained meshes with less use of memory
 - Updated `optixCurves` sample to include `--ribbon` option
 - Demonstrates use of new flat quadratic B-spline curve
- When debugging device code in OptiX shaders, you will need to set the environment variable `OPTIX_FORCE_DEPRECATED_LAUNCHER` to 1 prior to running the application.

What's new in 7.6.0

- New feature: Opacity MicroMaps (OMMs) allow RTCores to skip anyhit invocation for hardware accelerated triangles that have alpha masks. It uses a micromap defined on a barycentric grid to define opaque and transparent regions in the triangle. OMMs can accelerate scenes that make heavy use of alpha such as alpha cut out foliage, particle billboards and other places where alpha textures are used. In many cases, OMMs can replace the anyhit program entirely. See `optixCutouts` and `optixOpacityMicromap` samples for example usage. For more information please see section 5.11 Opacity Micromaps in the [programming guide](#).
 - OMMs are hardware accelerated on Ada Lovelace generation GPUs and also work efficiently on prior generations
 - OMMs can be overridden per instance using new fields defined in `OptixInstance`
 - See also the new header `optix_micromap.h` for helper functions related to OMMs
- Relocation API has been updated to facilitate relocation of OMM data that is part of geometry acceleration structures
 - The function `optixAccelCheckRelocationCompatibility` has been renamed to `optixCheckRelocationCompatibility`

- The struct `OptixAccelRelocationInfo` has been renamed to `OptixRelocationInfo`
- The parameters to `optixAccelRelocate` have been changed
 - Device pointers to the instance handles are no longer passed in. Instead, pass in an array of `OptixRelocateInput` structs, which can now include the device pointers to the instance handles.
 - Removed
 - `CUdeviceptr instanceTraversableHandles`
 - `size_t numInstanceTraversableHandles`
 - Added
 - `const OptixRelocateInput* relocateInputs`
 - `size_t numRelocateInputs`
 - Error code `OPTIX_ERROR_ACCEL_NOT_COMPATIBLE` has been renamed to `OPTIX_ERROR_NOT_COMPATIBLE`
- The OptiX Demand Loading library is no longer distributed with the OptiX SDK. It is now part of the OptiX Toolkit: <https://github.com/NVIDIA/optix-toolkit>.
- Symbols with weak linkage are now supported. Templated code that is not inlined can generate a function with weak linkage and would previously cause multiple defined symbol errors.
- `OPTIX_DEVICE_PROPERTY_LIMIT_MAX_SBT_RECORDS_PER_GAS` erroneously reported 2^{28} when the supported value is really 2^{24} . Applications using more than 2^{24} SBT records per GAS would see crashes.
- Fixed bugs when using curve type `OPTIX_PRIMITIVE_TYPE_ROUND_CATMULLROM`, `OPTIX_PRIMITIVE_TYPE_ROUND_CUBIC_BSPLINE`, or `OPTIX_PRIMITIVE_TYPE_ROUND_QUADRATIC_BSPLINE`:
 - The curve didn't return endcap intersections in case of endcaps "off" mode (`OPTIX_CURVE_ENDCAP_DEFAULT`)
 - The curve parameter was not set correctly when using `OPTIX_RAY_FLAG_TERMINATE_ON_FIRST_HIT`
- Addressed a number of warnings produced when using OptiX with Valgrind
- Report warnings when processing inline PTX when targeting OptiX IR
- New and updated SDK samples
 - New sample `optixOpacityMicromap`
 - Demonstrates the new Opacity Micromap feature for accelerating alpha cutouts
 - `optixCutouts` now uses Opacity Micromaps for some of the triangle mesh objects
 - New sample `optixConsole`
 - demonstrates how to build OptiX applications in Microsoft Visual Studio without the use of CMake
 - Prints to stdout an OptiX generated image converted to ASCII art
 - Deleted samples `optixDemandLoadSimple` and `optixDemandTexture`
 - Fix debug flags for makefile or other single configuration CMake generators
 - `optixPathTracer` now uses payload types to improve register pressure during ray traversal and to demonstrate the application of payload types

Known Issues

1. Pixel formats `OPTIX_PIXEL_FORMAT_UCHAR3` and `OPTIX_PIXEL_FORMAT_UCHAR4` are not supported by the Denoiser
2. Concurrent launches from the same pipeline will serialize automatically on the device
3. `OPTIX_COMPILE_DEBUG_LEVEL_FULL` support is still under development and may not generate correct debug information for `cuda-gdb` or `Nsight Compute VSE`

What's new in 7.5.0

- OptiX IR input for OptiX module creation
 - CUDA version 11.7 and above support generation of a proprietary binary format that conveys more information from NVCC to OptiX, including information required for debugging.
 - Invoke NVCC with `--optix-ir` to target this new format. Replacing `--ptx` with `--optix-ir` should be sufficient for most applications.
 - `optixModuleCreateFromPTX` will take this as input, similar to PTX input.
 - Inline PTX is still supported when using OptiX IR.
- Device code debugging preview
 - NSight Visual Studio Edition can show the value of variables inside of OptiX programs. In this preview version, some variables may not be present or have a value.
 - Only supported with OptiX IR input as described above.
 - Be sure to use `-G -O0` when compiling with NVCC to get debugging symbol information and use `OPTIX_COMPILE_DEBUG_LEVEL_FULL` and `OPTIX_COMPILE_OPTIMIZATION_LEVEL_0` in `OptixCompileOptions`.
 - A backtrace with only function names is still available with PTX input, similar to previous versions.
- New built-in sphere primitive
 - A geometry acceleration structure can now contain lists of spheres. Each sphere is specified by its center and radius.
 - On the host side, use the new build input type `OptixBuildInputSphereArray`. Similarly to curves, sphere intersection needs to be enabled by setting `OPTIX_PRIMITIVE_TYPE_FLAGS_SPHERE` in `OptixPipelineCompileOptions::usesPrimitiveTypeFlags`.
 - On the device side, sphere intersection returns one attribute. When the ray hits the sphere twice, the hit returned is the front face hit (entering the sphere), and the attribute contains the `t` value of the back face hit (exiting the sphere). Otherwise the attribute is 0. Note `optixIsFrontFaceHit` can be used to determine whether the hit is front face or back face.
 - The new device function `optixGetSphereData` can fetch any sphere's center and radius. Similarly to other get-data functions, it requires `OPTIX_BUILD_FLAG_ALLOW_RANDOM_VERTEX_ACCESS` to be set in `OptixAccelBuildOptions::buildFlags` before the acceleration structure is built. Setting this flag uses more memory.
 - See the SDK sample `optixSphere`.
- New error code `OPTIX_ERROR_DEVICE_OUT_OF_MEMORY`
 - Can be returned from previous asynchronous errors, or from `optixLaunch` when growing the stack memory required to support the launch.

- In previous driver versions, an `OPTIX_ERROR_UNKNOWN` would be returned under these circumstances.
- Memory allocation for stacks can be substantial when using large stacks; see `optixPipelineSetStackSize` to control this allocation.
- New `OptixGeometryFlags` enum
 - `OPTIX_GEOMETRY_FLAG_DISABLE_TRIANGLE_FACE_CULLING`
- Denoiser
 - A new mode combines 2X upscaling with denoising in a single operation. Use `OPTIX_DENOISER_MODEL_KIND_UPSCALE2X` or `OPTIX_DENOISER_MODEL_KIND_TEMPORAL_UPSCALE2X`.
 - Denoiser scratch memory requirements are now lower in some cases. The required scratch memory sizes for the API functions `optixDenoiserComputeIntensity` and `optixDenoiserComputeAverageColor` should now be queried with `optixDenoiserComputeMemoryResources` instead of relying on the equations that were previously provided in the function header documentation. For `optixDenoiserComputeIntensity` and `optixDenoiserComputeAverageColor`, the scratch memory requirements are significantly lower. The scratch memory size for these functions is stored in `OptixDenoiserSizes::computeAverageColorSizeInBytes` and `OptixDenoiserSizes::computeIntensitySizeInBytes`.
 - Temporal denoising now relies on an internal guide layer to carry information from the previous denoising pass, specified with `previousOutputInternalGuideLayer` and `outputInternalGuideLayer` in the `OptixDenoiserGuideLayer` struct. A set of guide layers must be provided to the denoising pass as both input and output. These layers have the format `OPTIX_PIXEL_FORMAT_INTERNAL_GUIDE_LAYER` and should be sized according to the field `internalGuideLayerPixelSizeInBytes` returned by `optixDenoiserComputeMemoryResources`. See the [programming guide](#) for more information.
 - `OptixDenoiserGuideLayer::previousOutputInternalGuideLayer` and `OptixDenoiserGuideLayer::outputInternalGuideLayer` must refer to two separate buffers, it is not possible to share memory between these buffers, regardless of tiling mode. After denoising a frame, these two buffers must be exchanged, so that `outputInternalGuideLayer` becomes `previousOutputInternalGuideLayer` for the next frame. Instead of copying data into the previous guide layer, it is recommended to use a double-buffering strategy by swapping the content of the two `OptixImage2D` structs. Also use double-buffering for `OptixDenoiserLayer::output` and `OptixDenoiserLayer::previousOutput`.
 - `OptixDenoiserParams::temporalModeUsePreviousLayers`
 - In temporal mode, setting this to 1 indicates the denoiser should read the values of the previous frame. Set to 0 for initial frames or when you want to reset the temporal sequence. In the first frame, when using temporal denoising modes, `OptixDenoiserGuideLayer::flow` must either contain valid motion vectors if available, otherwise the xy vectors must be set to zero (no motion). In temporal upscaling mode `OptixDenoiserLayer::previousOutput` is not accessed

- when `OptixDenoiserParams::temporalModeUsePreviousLayers` is not set.
- `OptixDenoiserSizes` struct has new fields
 - `computeAverageColorSizeInBytes`
 - `computeIntensitySizeInBytes`
 - `internalGuideLayerPixelSizeInBytes`
- `OptixDenoiserParams::denoiseAlpha` changed from a boolean flag to an enumerated type called `OptixDenoiserAlphaMode` which now has three possible values. These can be used for all denoiser models.
 - `OPTIX_DENOISER_ALPHA_MODE_COPY`: This is the default, alpha is copied from input to output and not denoised.
 - `OPTIX_DENOISER_ALPHA_MODE_ALPHA_AS_AOV`: This will apply the denoiser weights to the alpha channel, treating it similarly to other AOV channels. This is the most efficient method to get a denoised alpha channel.
 - `OPTIX_DENOISER_ALPHA_MODE_FULL_DENOISE_PASS`: A new denoising mode that is useful where the alpha channel has a substantially different noise profile than the RGB image. If artifacts appear, such as extra halo areas around alpha, this mode may yield better denoised alpha channels. However, this will execute a separate inference pass on alpha, meaning that denoise execution will take twice as long.
- When debug exceptions are enabled, invalid ray exceptions are thrown for rays with zero length direction (with denorm values treated as zero) and when `tmin` is negative.
- The many overloads of `optixTrace` have been replaced with a single templated function that can accommodate any number of payload values.
- Corrected documentation on the number of SBT records supported by a single GAS. The correct limit is 2^{24} . Improved the error message when this limit is exceeded.
- Compile time improvements
 - Improved concurrency when creating modules from PTX input in parallel. Improves compile times, and is especially noticeable when compiling many small modules.
- Fixed a bug where `optixAccelComputeMemoryUsage` and `optixAccelBuild` would change the current CUDA context.
 - Applications compiled against older versions of the NVIDIA OptiX SDK will see the context remain the same after the calls.
 - Applications compiled against 7.5 are required to have the current CUDA context be the context associated with the `OptixDeviceContext` when calling `optixAccelBuild` or any other OptiX function that invokes work on the GPU. For performance, the application should minimize the number of CUDA context changes around OptiX calls.
 - Validation mode will now catch when the CUDA context is incorrect. Undefined behavior or errors may result otherwise.
- Fixed a bug that sometimes led to a multiply defined symbol error when linking a pipeline with multiple built-in IS modules.

What's new in 7.4.0

- New compiler backend
 - Supports parallel compilation and other new features

- This change should be invisible to applications, but if you have performance regressions, unexpected bugs or error messages that look like "New backend is missing implementation for PTX intrinsic <XYZ>", please contact us.
- Parallel compilation within a module exposed through new API
 - Exposes parallelism within a single module and between modules
 - Flexible threading model that can be used within existing application work queues
 - New API functions
 - `optixModuleCreateFromPTXWithTasks`
 - `optixModuleGetCompilationState`
 - `optixTaskExecute`
 - The following functions can now return `OPTIX_ERROR_ILLEGAL_DURING_TASK_EXECUTE`
 - `optixTaskExecute` if the task is already being executed on another thread
 - `optixModuleDestroy` if a task associated with the module is currently being executed
 - Added sample `optixCompileWithTasks` and SDK header `CompileWithTasks.h` that demonstrate this feature
- Payload types
 - To reduce register consumption in complex pipelines, OptiX 7.4 has introduced a mechanism to annotate the usage of each payload register. OptiX will analyze the lifetimes and reuse registers where possible.
 - The maximum number of payload registers has been increased from 8 to 32
 - These two features allow more workloads to improve performance by passing payload values in registers instead of local memory
 - Instead of supplying the number of payload registers in `OptixPipelineCompileOptions`, payload utilization information is now passed through `OptixModuleCompileOptions`. This contains an array of `OptixPayloadType` objects that annotate each payload value with their READ or WRITE usage.
 - The original mechanism of passing the number of payload registers in `OptixPipelineCompileOptions` will continue to work, though all modules in the pipeline must use the same mechanism
 - Refer to the programming guide for additional information
- Catmull-Rom round curve primitive type added
 - Same data layout as round quadratic b-splines
 - Access with new primitive enum values
 - `OptixPrimitiveType::OPTIX_PRIMITIVE_TYPE_ROUND_CATMULLROM`
 - `OptixPrimitiveTypeFlags::OPTIX_PRIMITIVE_TYPE_FLAGS_ROUND_CATMULLROM`
 - Control points can be retrieved on the device using `optixGetCatmullRomVertexData`
- Selectable endcap behavior for quadratic and cubic curves added, including the new Catmull-Rom curve
 - `OptixCurveEndcapFlags` used in `OptixBuildInputCurveArray::endcapFlags` and `OptixBuiltinISOptions::curveEndcapFlags`
 - Default is endcaps OFF. Previously the endcap behavior was ON.

- `OptixBuiltinISOptions` now has a `buildFlags` parameter that should match the ones supplied to `OptixAccelBuildOptions` when building the AS containing the built-in primitives
- When building acceleration structures containing curves with default build flags, an average of 7% reduction in memory was observed across a suite of datasets
- Demand Loading
 - An eviction mechanism now allows texture tiles to be reused when device memory is limited, allowing larger scenes to be rendered
 - Texture lookup can be skipped when texture gradients are large, using an optional base color that can be specified when the texture is initialized (either from metadata or from the coarsest miplevel)
 - Small textures (32x32 or less) no longer employ CUDA sparse textures, which saves memory and improves performance
 - UDIM UV mapping is now supported. A texture atlas can be constructed from a collection of sparse textures, each of which can leverage the small texture and base color optimizations.
 - API changes:
 - The `EXRReader` class has moved to a separate `ImageReader` library, with its own namespace
 - `createDemandLoader` no longer requires a vector of active device ordinals
- Denoiser
 - Added `OptixDenoiserModelKind::OPTIX_DENOISER_MODEL_KIND_TEMPORAL_AOV`
 - When tiling the denoiser the overlap is returned in the API, but you may notice the value has increased from 64 to 128
- `OptixCompileDebugLevel`
 - Removed `OPTIX_COMPILE_DEBUG_LEVEL_LINEINFO`
 - Added
 - `OPTIX_COMPILE_DEBUG_LEVEL_MINIMAL` - generate information that does not impact performance (currently generates line information useful for profiling)
 - `OPTIX_COMPILE_DEBUG_LEVEL_MODERATE` - generate some debug information with slight performance costs (currently generates a debug frame necessary to show a backtrace)
- Removed `OptixInstanceFlags::OPTIX_INSTANCE_FLAG_DISABLE_TRANSFORM`. This flag did not operate as intended, and users were already required to provide an identity instance transform.
- New Samples
 - `optixOpticalFlow`: demonstrates the use of the NVIDIA Optical Flow SDK to provide motion vectors for temporal denoising in lieu of application-provided motion vectors.
 - `optixCompileWithTasks`: demonstrates the use of the new parallel compile interface and contains a simple interface that can be adapted to many applications.
- Improved compile caching behavior with many concurrent accesses. If the database was locked, it would previously disable the cache.
- Name of the compilation database file has been changed to "optix7cache.db". The directory location has not changed.
- New warning issued when creating an `OptixDeviceContext` when the device memory size exceeds the host memory size
- PTX input that calls `__assertfail` no longer causes linking errors

- Launch parameter specialization now allows for single byte specialization regardless of whether the byte is part of a larger element such as int or float. This should reduce false errors when using specialization.
- Note: For OptiX 6 applications
 - Setting `RT_GLOBAL_ATTRIBUTE_ENABLE_RTX` to 0 via `rtGlobalSetAttribute` will result in an error as RTX is now the only supported mode for OptiX 6. The legacy megakernel execution strategy is no longer supported.
 - Name of the compilation database file has been changed to "optixcache.db", but the directory location has not.

What's new in 7.3.0

- Denoiser for temporal images and API simplifications
 - Temporal denoising is now supported with a new built-in model kind. In this mode a sequence of images can be denoised. The AI network was trained to reduce flickering for camera or geometry animations. It requires the denoised beauty image from the previous frame as input as well as flow (motion) vectors. Currently temporal denoising is not supported for AOVs.
 - Enable by using the new `OPTIX_DENOISER_MODEL_KIND_TEMPORAL` enum value from `OptixDenoiserModelKind`.
 - `optixDenoiserCreate` and `optixDenoiserSetModel` have been merged. `optixDenoiserCreate` now uses an enum to select the built-in model kind and `optixDenoiserCreateWithUserModel` that takes a user model.
 - `optixDenoiserInvoke`'s input has changed to specify guide layers specified explicitly in `OptixDenoiserGuideLayer` (e.g. albedo, normal or pixel flow layers) and separately from beauty layers which are now specified in `OptixDenoiserLayer`. `optixUtilDenoiserInvokeTiled` has similarly been updated to the new interface.
 - `OptixDenoiserOptions` changed to specify whether albedo and normal guide layers will be provided during `optixDenoiserInvoke`
 - New `OptixPixelFormat`
 - `OPTIX_PIXEL_FORMAT_HALF2`
 - `OPTIX_PIXEL_FORMAT_FLOAT2`
- Demand Loading
 - The `launchPrepare` and `processRequests` methods are now asynchronous, taking a CUDA stream argument on which operations are enqueued.
 - Requests for sparse texture tiles are now processed in the background by multiple CPU threads. This greatly improves concurrency, keeping the CPU busy with I/O and texture decompression while OptiX kernels perform rendering work on GPU.
 - Multiple streams are now supported. A degree of latency hiding can be accomplished by rendering a framebuffer as multiple tiles in round-robin fashion: while the texture data for one framebuffer tile is loading, work can proceed on subsequent tiles, returning to the first when its data is ready.
 - NVIDIA OptiX now provides the following texture footprint functions, which are used by the NVIDIA OptiX Demand Loading library to quickly determine which sparse texture tiles are

required. These functions are hardware-accelerated on Turing and Ampere GPUs, with software emulation on older architectures.

- `optixTexFootprint2D`
- `optixTexFootprint2DLod`
- `optixTexFootprint2DGrad`
- Improved curve intersectors
 - o There is a new faster intersector for cubic and quadratic curves (based on Reshetov's [Phantom intersector](#)).
 - o There is a new intersector for piecewise linear curves that is slightly faster and higher precision.
 - o The new curve intersectors cull backfaces, treating the swept curve primitives as hollow. Rays originating inside the primitive will now exit the primitive.
 - o Caveat: Internal rays can still hit internal end caps (invisible end caps between segments of a strand).
 - o A bug in compaction of acceleration structures with curves has been fixed.
- New SDK samples:
 - o `optixDemandLoadSimple` - Demonstrates a simple use of the Demand Loading library.
 - o `optixModuleCreateAbort` - Compiles modules in separate processes, which can be interrupted.
 - o `optixMotionGeometry` - Demonstrates motion blur for vertex positions, SRT transforms, matrix transforms, and combined motion.
 - o `optixVolumeViewer` - Demonstrates incorporating OpenVDB volumes in an OptiX render, by using the open source [NanoVDB](#) library.
- Improved SDK samples:
 - o `optixDenoiser`
 - New mode for temporal denoising
 - Tiling operation now supported.
 - o `optixDemandTexture` - Now demonstrates the use of multiple streams for demand loading.
- `OptixPipelineCompileOptions` struct has changed. Please make sure to zero initialize this struct to avoid compilation errors.
 - o `OptixPipelineCompileOptions pipelineCompileOptions = {};`
- Validation mode now checks the stream state before executing API functions that take a stream.
- New device function `optixGetInstanceTraversableFromIAS`
 - o Return the traversable handle of a given instance in an Instance Acceleration Structure (IAS). This handle is not directly traversable, but can be used to query instance transforms and other information.
- New `OptixBuildFlag` `OPTIX_BUILD_FLAG_ALLOW_RANDOM_INSTANCE_ACCESS`. This flag is required to call `optixGetInstanceTraversableFromIAS`.
- New device function `optixGetInstanceChildFromHandle`
 - o Returns child traversable handle from an `OptixInstance` traversable
- New exception codes that can be triggered by `optixGetTriangleVertexData` and `optixGetInstanceTraversableFromIAS` when exceptions are enabled.
 - o `OPTIX_EXCEPTION_CODE_INVALID_VALUE_ARGUMENT_0`
 - The value passed in `ias` to `optixGetInstanceTraversableFromIAS` is not a valid Instance AS.

- The value passed in `gas` to `optixGetTriangleVertexData` is not a valid Geometry AS.
- `OPTIX_EXCEPTION_CODE_INVALID_VALUE_ARGUMENT_1`
 - The index passed is out of range.
- `OPTIX_EXCEPTION_CODE_INVALID_VALUE_ARGUMENT_2`
 - The `sbtGASIndex` passed to `optixGetTriangleVertexData` is out of range.
- `OPTIX_EXCEPTION_CODE_UNSUPPORTED_DATA_ACCESS`
 - `optixGetTriangleVertexData` was called on an acceleration structure that was built without `OPTIX_BUILD_FLAG_ALLOW_RANDOM_VERTEX_ACCESS` set.
 - `optixGetInstanceTraversableFromIAS` was called on an acceleration structure built without `OPTIX_BUILD_FLAG_ALLOW_RANDOM_INSTANCE_ACCESS` set.
 - An acceleration structure built with motion was used in a pipeline without motion enabled.
- If the OptiX compile cache is corrupted, OptiX will now attempt to delete and reinitialize the cache.
- Added `OPTIX_CACHE_MAXSIZE` environment variable to control the size of the disk cache.
- AS builds have been optimized for faster trace times.
- PTX must be recompiled with the new SDK headers if the host uses the new SDK.
- NVRTC is no longer enabled by default when compiling the SDK samples. It can be enabled by setting the CMake variable `CUDA_NVRTC_ENABLED`.
- Bug fixes:
 - Fixed bug to allow `optix_stack_size.h` to be able to be included in more than one compilation unit.
 - Fixed bug with `surf2Dwrite` calls having no effect.
 - Fixed bug with some denormal floats being treated as zero.
 - Fixed some bugs that would prevent correct line information from being used in profiling and debugging.

What's New in 7.2.0

- Specialization is a powerful new feature that allows renderers to maintain generality while increasing performance on specific use cases. A single version of the PTX can be supplied to OptiX and specialized to toggle specific features on and off. The OptiX compiler is leveraged to fold constant values and elide complex code that is not required by a particular scene setup. Specialized values are supplied during module creation with `OptixModuleCompileOptions::boundValues`. See the Programming Guide section 6.3.1, "Parameter specialization", and the `optixBoundValues` sample.
- Demand loading source library
 - Enables textures to be loaded on demand, which greatly reduces memory requirements, start-up time, and disk I/O compared to preloading textures.
 - Requires the CUDA 11.1 toolkit.
 - See section 14 "Demand-loaded sparse textures" of the NVIDIA OptiX Programming Guide for a detailed technical introduction.
 - The `optixDemandTexture` sample demonstrates how to use this library.

- This library supersedes the lower-level `optixPaging` library. The `optixDemandPaging` sample shows how to use `optixPaging` directly.
 - Known issue: wrap mode and mirror mode are not yet supported for CUDA sparse textures.
- There is a new mode in the denoiser that uses a neural network to predict a filter kernel instead of the final image. The filter weights can be applied to multiple layers or AOVs with just an incremental cost.
 - To select this mode, use `OptixDenoiserModelKind::OPTIX_DENOISER_MODEL_KIND_AOV`
 - Additional quality may be achieved by computing the average color using a new API function `optixDenoiserComputeAverageColor` and supplying the value to the denoiser using `OptixDenoiserParams::hdrAverageColor`
- To aid in debugging there is a new validation mode that runs additional checks during runtime and enables all debug exceptions.
 - Enable validation mode with: `OptixDeviceContextOptions::validationMode`
 - New error code when validation catches an error: `OPTIX_ERROR_VALIDATION_FAILURE`
 - APIs that take `CUstream` arguments are synchronized on the stream to check for errors before proceeding.
 - `optixLaunch` will synchronize after the launch and report errors
 - If any OptiX debug exceptions were thrown during launch, a CUDA launch error is triggered to prevent proceeding.
 - Validation mode reduces performance. Remember to turn it off.
- New debug exceptions
 - `OPTIX_EXCEPTION_CODE_CALLABLE_INVALID_SBT`
 - The callable program SBT record index was out of bounds
 - `OPTIX_EXCEPTION_CODE_CALLABLE_NO_DC_SBT_RECORD`
 - The callable program SBT record does not contain a direct callable program
 - `OPTIX_EXCEPTION_CODE_CALLABLE_NO_CC_SBT_RECORD`
 - The callable program SBT record does not contain a continuation callable program
- When linking, unresolved and multiply defined symbols will produce detailed error messages in the logs.
- Instance bounds are now computed automatically, even with motion and motion transforms. Instance acceleration structure build inputs no longer have the optional `OptixBuildInputInstanceArray::aabbs`.
 - Note OptiX will compute AABBs for all applications running against driver 455+ regardless of the version of the SDK it was built against. OptiX will ignore the AABBs supplied by applications compiled with earlier SDKs.
- You can now unload the NVIDIA OptiX driver DLL or DSO. See `optixUninitWithHandle()` in the `optix_stubs.h` header file.
 - If other threads in the process have a handle to the DLL it will continue to be loaded until the last handle is released.

- There is a new error code when `optixUninitWithHandle` fails:
`OPTIX_ERROR_LIBRARY_UNLOAD_FAILURE`
- Fixed a bug where PTX compiled with `-lineinfo` in CUDA 11 could cause errors when loading into OptiX.
- Fixed bugs related to large numbers of curves.
- Optimized traversal when only triangle geometry is enabled with single level instancing, that is, `OptixPipelineCompileOptions::usesPrimitiveTypeFlags` equals exactly `OPTIX_PRIMITIVE_TYPE_FLAGS_TRIANGLE` and `OptixPipelineCompileOptions::traversableGraphFlags` equals exactly `OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_LEVEL_INSTANCING`
- SDK
 - By default the SDK will target PTX compilation for SM 60 (Volta+). Set CMake variables `CUDA_NVCC_FLAGS` and `CUDA_NVRTC_FLAGS` to use an older SM version if desired.
 - New samples
 - `optixBoundValues`
 - `optixDemandTexture`
 - `optixDenoiser`
 - `optixDynamicGeometry`

What's New in 7.1.0

- Added curves as a new type of geometric primitive. Curves are swept surfaces used to represent long thin strands, such as for hair, fur, or cloth fibers. Linear, quadratic, and cubic B-spline bases are supported. Motion blur is supported.
 - Added new GAS build input type for curves. See the NVIDIA OptiX Programming Guide section 5.2, “Curve build inputs”.
 - Hit programs can access the curve parameter value at the hit point, and the curve’s geometric data which is stored in the acceleration structure (GAS). Utility code is provided in the SDK (`cuda/curve.h`) to compute the curve surface position, tangent, and normal.
 - Each SBT program group for curves requires a built-in curves intersection program, returned by the new host function `optixBuiltinISModuleGet`. Motion blur for curves is enabled [here](#).
 - Pipelines can now indicate which primitive types they support via `OptixPipelineCompileOptions::usesPrimitiveTypeFlags`. If your scene contains curves, they must be enabled here. If your scene geometry is all triangles (no curves and no custom primitives), set these flags to enable only triangles, for optimal performance.
 - In hit programs, the recommended method to discriminate among hit types is to use the new methods `optixGetPrimitiveType`, `optixIsFrontFaceHit`, and `optixIsBackFaceHit`. The old methods, e.g. `optixIsTriangleFrontFaceHit`, still work.
 - See the NVIDIA OptiX Programming Guide section 8, “Curves”, for additional information.
- The denoiser has several improvements for quality and performance in addition to some API changes.

- Added support for `OptixDenoiserInputKind::OPTIX_DENOISER_INPUT_RGB_ALBEDO_NORMAL`.
- Added support for tiling. See `optix_denoiser_tiling.h` in the SDK for helpers to use tiling.
- `OptixDenoiserOptions::pixelFormat` has been removed, because it is no longer needed.
- `OptixDenoiserSizes::minimumScratchSizeInBytes` and `recommendedScratchSizeInBytes` have been removed and replaced with `withOverlapScratchSizeInBytes` and `withoutOverlapScratchSizeInBytes`.
- Increase instancing limits. Query limit with `OPTIX_DEVICE_PROPERTY_LIMIT_MAX_INSTANCES_PER_IAS`. The limit has been increased to 2^{28} instead of 2^{24} . `OptixInstance::sbtOffset` now also supports values up to 2^{28} .
- Removed `OptixPipelineLinkOptions::overrideUsesMotionBlur`. This option is no longer needed.
- Added `OptixTransformFormat` used in `OptixBuildInputTriangleArray::transformFormat`. Allows to specify the format of `OptixBuildInputTriangleArray::preTransform`. Value should be `OPTIX_TRANSFORM_FORMAT_MATRIX_FLOAT12` when `preTransform` will be supplied to the build and `OPTIX_TRANSFORM_FORMAT_NONE` when `preTransform` is unused. A nullptr can now be used for `optixAccelComputeMemoryUsage` instead of supplying a valid or dummy pointer.
- Several new device exceptions were added to catch common errors. They are active when debug exceptions are enabled in `OptixPipelineCompileOptions::exceptionFlags`.
 - Invalid ray exception
 - Checks for NaN and Inf in the ray parameters passed `optixTrace`
 - New exception code: `OPTIX_EXCEPTION_CODE_INVALID_RAY`
 - `optixGetExceptionInvalidRay` returns details of the exception in a struct called `OptixInvalidRayExceptionDetails`
 - Callable program parameter mismatch
 - Currently only checks the number of parameters and not the types, since PTX can lose type information.
 - New exception code: `OPTIX_EXCEPTION_CODE_CALLABLE_PARAMETER_MISMATCH`
 - `optixGetExceptionParameterMismatch` returns details of the exception in a struct called `OptixParameterMismatchExceptionDetails`
 - Ensure that the built-in intersection program assigned to the SBT matches the GAS
 - New exception code: `OPTIX_EXCEPTION_CODE_BUILTIN_IS_MISMATCH`
 - Ensure that when `optixTrace` is called with a single level GAS as the trace target that `OptixPipelineCompileOptions::traversableGraphFlags` is set to either `OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_SINGLE_GAS` or `OPTIX_TRAVERSABLE_GRAPH_FLAG_ALLOW_ANY`.
 - new exception code: `OPTIX_EXCEPTION_CODE_UNSUPPORTED_SINGLE_LEVEL_GAS`

- The shader encountered an unsupported primitive type when calling `optixGetLinearCurveVertexData`, `optixGetQuadraticBSplineVertexData`, `optixGetCubicBSplineVertexData`, or `optixGetPrimitiveType`. Supported primitive types are set with `OptixPipelineCompileOptions::usesPrimitiveTypeFlags`.
 - New exception code: `OPTIX_EXCEPTION_CODE_UNSUPPORTED_PRIMITIVE_TYPE`
- When `OPTIX_EXCEPTION_CODE_TRAVERSAL_INVALID_HIT_SBT` is thrown, `optixGetPrimitiveIndex` is no longer supported. `optixGetSbtGASIndex` should be used instead. See `optixDumpExceptionDetails` for details.
- `optixThrowException` is now elided when user exceptions are disabled instead of producing an error.
- Added `optixGetExceptionLineInfo` device function accessible in exception programs.
 - Returns a string that includes information about the source location that caused the current exception. Only supported for certain exceptions, and requires line information in the PTX (`--lineinfo`) and a debug level that supports line info (`OPTIX_COMPILE_DEBUG_LEVEL_LINEINFO` and `OPTIX_COMPILE_DEBUG_LEVEL_FULL`).
- Change `OptixCompileOptimizationLevel` and `OptixCompileDebugLevel` enum values.
 - Added a `DEFAULT` value of 0 (used when zero initializing the structs).
 - `OptixCompileOptimizationLevel::OPTIX_COMPILE_OPTIMIZATION_DEFAULT` continues to mean `OPTIMIZATION_LEVEL_3`.
 - `OptixCompileDebugLevel::OPTIX_COMPILE_DEBUG_LEVEL_DEFAULT` (new) adds lineinfo (same as `OPTIX_COMPILE_DEBUG_LEVEL_LINEINFO`).
- Fixed support in optix headers for cuda runtime compilation using `nVRTC`.
- Enable compaction support for acceleration structures on non-RTX GPUs.
- OptiX will attempt to reset a corrupted compile disk cache.
- Added `OptixIndicesFormat::OPTIX_INDICES_FORMAT_NONE` which must be used when vertex indices are not present (e.g. triangle soup).
- Added `OptixVertexFormat::OPTIX_VERTEX_FORMAT_NONE` for use when initializing `OptixBuildInputTriangleArray::vertexFormat`. The value, `vertexFormat`, must be set to something other than `OPTIX_VERTEX_FORMAT_NONE` before `OptixBuildInputTriangleArray` can be used.
- Additional checks are now in place when calling `optixPipelineSetStackSize`. If the `maxTraversableGraphDepth` is greater than maximum supported by the device an error is generated. `maxTraversableGraphDepth` also must be greater than zero. `maxTraversableGraphDepth` must also be compatible with `OptixPipelineCompileOptions::traversableGraphFlags`.
- When used with Nsight Compute, `optixLaunch` and `optixAccelBuild` are now marked in the timeline.
- Very large AABBs are now clamped to +/- 2^40 for non-motion acceleration structures.
- `optixGetTriangleVertexData` support has been extended to all supported GPUs.

- `optixGetTriangleVertexData` support has been fixed for large meshes.
- Fixed a crash on non-RTX GPUs when rendering with a refit IAS.
- New SDK samples.
 - `optixCallablePrograms` - Demonstration of *callable program* usage and SBT setup.
 - `optixCurves` - Minimal curve API usage example.
 - `optixDenoiser` - Demonstration of NVIDIA OptiX AI Denoiser usage.
 - `optixDynamicGeometry` - Shows typical setup for changing geometry between frames.
 - `optixHair` - More complicated curve API example.
 - `optixNVLink` - Demonstrates NVLink usage within an OptiX application.

What's New in 7.0

- 7.0 introduced the NVIDIA OptiX 7 API, a new low-level CUDA-centric API giving application developers direct control of memory, compilation, and launches while maintaining the programming model and shader types from previous versions of OptiX.
- Minimal host state is maintained. Scene graphs, materials, etc., are managed by the application rather than by OptiX.
- GPU memory is managed by the application using CUDA. (No OptiX buffers or variables)
- GPU launches are explicit and asynchronous using CUDA streams.
- Shader compilation is explicit. (Similar to DXR or Vulkan)
- All host functions are thread-safe.
- Source code for demand loading library is included and designed for direct inclusion in production applications.
- Multi-GPU operation is managed by the application.